

AI-Enabled Full Stack Developer Roadmap & Resources

This roadmap is designed for a frontend developer transitioning into an AI-enabled Full Stack Engineer. It includes the best resources, learning order, and project ideas for becoming future-ready in the AI industry.

Phase 1 — Frontend Foundation

React / Next.js

- Next.js Docs — <https://nextjs.org/docs>
- Focus on App Router, Server Components, Data Fetching, and Authentication

TypeScript

- TypeScript Handbook — <https://www.typescriptlang.org/docs/>
- Learn interfaces, generics, utility types, and API typing

Styling

- Tailwind CSS — <https://tailwindcss.com/docs>
- shadcn/ui — <https://ui.shadcn.com>

Phase 2 — Backend + APIs

Node.js + Express

- Node.js Docs — <https://nodejs.org/en/docs>
- Express.js Docs — <https://expressjs.com>

Database + SQL

- SQLBolt — <https://sqlbolt.com>
- LeetCode SQL — <https://leetcode.com/problemset/database/>
- PostgreSQL Docs — <https://www.postgresql.org/docs/>
- Prisma ORM — <https://www.prisma.io/docs>

Backend Topics

- Authentication
- JWT
- Validation
- Rate Limiting
- RBAC
- Pagination

- Error Handling

Phase 3 — AI / LLM Development

OpenAI APIs

- OpenAI Docs — <https://platform.openai.com/docs>
- Learn chat completions, streaming, embeddings, and function calling

Vercel AI SDK

- Docs — <https://sdk.vercel.ai/docs>
- Learn streaming UI and AI chat interfaces

RAG Systems

- LangChain — <https://python.langchain.com/docs/introduction/>
- LlamaIndex — <https://docs.llamaindex.ai>
- Pinecone — <https://www.pinecone.io/learn/>
- ChromaDB — <https://docs.trychroma.com>

AI Agents

- LangGraph — <https://langchain-ai.github.io/langgraph/>
- CrewAI — <https://docs.crewai.com>

Phase 4 — Python for AI

Python

- Python Tutorial — <https://docs.python.org/3/tutorial/>

FastAPI

- FastAPI Docs — <https://fastapi.tiangolo.com>

Focus Areas

- Async Python
- Automation
- AI Pipelines
- APIs
- File Handling

Phase 5 — AWS + Deployment

AWS

- AWS Skill Builder — <https://skillbuilder.aws>
- AWS Docs — <https://docs.aws.amazon.com>

Docker + CI/CD

- Docker Docs — <https://docs.docker.com>
- GitHub Actions — <https://docs.github.com/actions>

Git

- Git Docs — <https://git-scm.com/doc>
- Learn Git Branching — <https://learngitbranching.js.org>

Best YouTube Channels

Frontend + AI

- Codevolution
- Jack Herrington
- Web Dev Simplified
- AI Jason
- Cole Medin
- Sam Witteveen

System Design

- ByteByteGo
- CodeKarle

Best Portfolio Projects

Projects

- AI Chat App
- AI PDF Chat (RAG)
- AI Analytics Dashboard
- AI-Powered QR Menu SaaS

Final Advice

Key Strategy

- Learn → Build → Improve → Deploy

- Focus on deep projects instead of many shallow tutorials
- Become an AI-enabled Full Stack Engineer